1495 UNIVERSITY OF ABERDEEN | CELEBRATING 525 YEARS 1495 – 2020

ABERDEEN 2040

Introduction

JC3503: Data Mining & Visualisation

Lecture 1

First off...

Welcome to JC3503: Data Mining and Visualisation!

In this course, you will learn the skills to analyse, visualise, and understand vast quantities of data.

We will discuss some of the approaches and methods for doing so, alongside some of the wider considerations within our data-driven society!

Today...

- About the course
- Learning outcomes
- Course schedule
- Assessments

Data Mining & Visualisation

Broader Context & Background

ABERDEEN 2040

Data is Ubiquitous

In the modern world, data is everywhere...

… and vast quantities of data are being generated every day!

Social Data

Think about how much data a single person might generate:

- Messages
- Photos
- Location / GPS
- Online (and offline) purchasing
- Media consumption (music, films, websites visited)
- Social media content

Technical Data

Think about the data being generated by technical systems:

- Online services
- Banking systems
- Healthcare / medical devices
- Phones and computers
- Smart homes / IoT interactions
- Modern vehicles

Historic Data

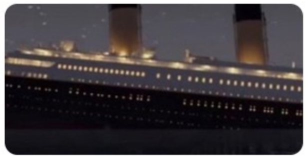
Think about how much historic data is available!

KAGGLE · GETTING STARTED PREDICTION COMPETITION · ONGOING

Submit Prediction

Titanic - Machine Learning from Disaster

Start here! Predict survival on the Titanic and get familiar with ML basics



	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Data Analysis

Data is everywhere, and there is far too much for anyone to process and understand it themselves.

Modern computers give us the ability to perform sophisticated analyses on vast amounts of data, very quickly.

In this course, you will learn some of the skills to do this.

In short, the aim is for you to learn to think like a data scientist!

Data Mining & Visualisation

Data Mining is the process of discovering patterns and extracting useful information from data.

Data Visualisation is the process of designing visual representations of data.

Fundamentally, both of these topics are about **understanding your data!**

Further Reading

Data Mining: Practical Machine Learning Tools and Techniques
Ian H. Witten & others (4th Edition, Morgan Kaufmann, 2017).

The Data Science Design Manual
Steven S. Skiena (Springer, 2017).

Data Visualisation: A Handbook for Data Driven Design
Andy Kirk (2nd edition, SAGE, 2019).

Learning Outcomes & Course Schedule

Learning Outcomes of JC3503

1. Manipulate, format, prepare, and clean data sets prior to analysis

2. Analyse complex datasets by applying data pre-processing, exploration, clustering and classification, time series analysis, and others

3. Design appropriate visualisation solutions for different applications, scenarios, and audiences

Technical Tools used in JC3503

We will use Python and Jupyter for the labs and assignment:

- Pandas for dataframes
- Seaborn for visualisations (and datasets)
- Scipy
- Sklearn

But our learning outcomes are language agnostic.

Other languages/packages/tools are widely used

Course Structure

- Week 1–4 (Part 1 & 2 of course)
  - 4 lectures of course material per week
  - 2 practical labs (double sessions) per week
- Week 5–8 (Part 3 of course)
  - 2 lectures of course material per week
  - 1 practical labs (double sessions) per week
- Week 9–12 (Part 4 of course)
  - 2 lectures of course material per week
  - 1 practical labs (double sessions) per week

Part 1: Overview of Data Mining

- Introduction to Data Mining
- Exploratory Data Analysis (EDA)
- Data Visualisation
- A/B Testing and Null Hypothesis Statistical Testing (NHST)

Part 2: Supervised Learning

- Regression and Classification
- Decision Trees
- Naïve Bayes
- Support Vector Machines

Part 3: Unsupervised Learning

- Clustering
- Data Dimensionality
- Association Rule Mining

Part 4: Mining Different Types of Data

- Time Series
- Text Mining

Learning Assessments

Assignment and Exam

Learning Assessments

- Written Exam (75% of course grade)
  - Will cover all lectures
  - Calculator not permitted
  - Exact date and time TBC

- Practical Assignment (25% of course grade)
  - Individual work only (no group work permitted)
  - No generative AI use permitted
  - Assignment due after week 12

Exam Information

As part of this course, you will have a written exam, which will be worth 75% of your course grade.

Half of the exam will consist of multiple choice questions, across all of the course's content.

The remaining half of the exam will involve larger questions, which will require more effort to answer.

A calculator is not permitted in the exam hall, so all required calculations will be possible without one.

Exam Information – Past Papers

A number of past papers will be made available to you via MyAberdeen, to help you revise and practice for the exam.

Alongside the practicals, lab sessions are a great opportunity to prepare for the exam, and raise any questions you have.

Note: some older past papers might include questions that are no longer covered by this course. If you're not sure whether you should know something, <u>ask during the labs</u>.

Assignment Information

As part of this course, you will also be required to demonstrate your ability to analyse an assigned dataset.

Importantly, much like data mining itself, this is an exploratory task; there is no clear end-goal or natural end-point.

You will be expected to use your analysis to demonstrate your understanding of this course in any way that you see fit.

Assignment Information – Objective

In other words, you have to determine what, how, and why you undertake particular data mining techniques.

You need to determine what level of data analysis is appropriate, given the scale and the timing of the assignment.

This course will cover a broad range of approaches to try, and you should explore your data to guide your analysis.

Assignment Information – Objective

There are many ways to explore and analyse this dataset. We want to see your **processes** and **justifications** for any analysis that you do.

Markdown should be used to explain what you are doing (at a high-level), and to add your own interpretation of what the data and the results are showing, at each step of your analysis.

Document your code well, so that we can follow along with your thinking for undertaking particular analyses or processes.

<u>**All code, comments, and markdown must be in English.**</u>

Assignment Information – Marking

The aim of this assignment is to demonstrate:

- your ability to undertake exploratory data analysis on new data.
- your depth and breadth of knowledge with relation to data mining and visualization, and the broader course.
- your communication skills (clear, technical contents and sound reasoning).

Assignment Information – Marking

The assignment is graded out of 100 marks:

- Appropriate use of EDA and descriptive statistics (~30 marks)
- Appropriate use of data mining techniques (~30 marks)
- Appropriate use of data visualization (~20 marks)
- Appropriate documentation via code comments and markdown (~20 marks)

In reality, grading will be more flexible with mark allocation, but this gives you some idea of how you should be prioritizing

Assignment Information – Instructions

The dataset, a .ipynb notebook template, and an information document (outlining all expectations of the assignment) will be made available on MyAberdeen soon. **<u>Make sure you read this carefully</u>.**

You will download the dataset and the notebook (template). All of your analysis should be included within this notebook.

Make sure that your name, email address, and student ID are all included within the markdown at the top of the notebook.

Assignment Information – Submission

To submit your completed notebook, you should rename the notebook to your student ID (e.g. ‘50080001.ipynb’), and submit to MyAberdeen before the deadline.

You should **not** zip this file, include any other files with your submission, or add anything else to the filename.

Submitting anything else, or not following this submission process may lead to points deductions. If you’re not clear, then ask!

Assignment Information – Plagiarism & Group Working

<u>Assignments are to be undertaken individually</u>:

- Do not share code/scripts, or discuss the steps you have taken with your fellow coursemates.
- Do not obtain analysis scripts or code blocks online (without clearly outlining what is obtained and attributing the source)
- The university plagiarism policy applies:
    - Assignments will be checked for plagiarism, between students and across various online resources/repositories.

Assignment Information – Generative AI Use

Generative AI is not permitted for use in this assignment (Level 0):

- This includes using AI to generate code, assist your analysis, translate text to English, or any other uses of generative AI.
- Suspected use may result in the academic misconduct process being initiated.
- See MyAberdeen and the University's guidance on Generative AI use.

Assignment Information – Tip #1

Important: This assignment is not an ML challenge – Don't just focus on *prediction for prediction's sake*.

Your main objective is towards inference and understanding of the dataset's variables (and their relationships).

Simply training ML models for the sake of prediction will not result in a good mark.

Assignment Information – Tip #2

The dataset will have many interesting attributes, including categorical, quantitative, and text-based variables.

Don’t just focus on one or two. Explore them! Use them to understand the broader patterns and trends within the data!

<u>We are looking for curiosity and creativity</u>!

Assignment Information – Tip #3

An instructional information sheet will be uploaded to MyAberdeen, with rules for submitting. Make sure you follow these closely!

Don’t lose marks for not following the correct process:

E.g. submitting a notebook that is not named correctly, zipping the notebook before uploading, including non-English comments/code, etc.

If in doubt, ask!

Assignment Information – Extensions

You should apply for an extension by emailing your request (using your UoA email address) to: **uoa-ji-enquiries@abdn.ac.uk**

Students should also include any supporting evidence where possible e.g medical letter.

Please read the Extensions and Late Submission of Work policy for full details. This is linked in the information sheet on MyAberdeen.

Assignment Information – Late Penalties

If work has been submitted late without an extension (or beyond the agreed extension date) penalties will be applied in the following way:

- For coursework submitted up to 24 hours late, **we will deduct 2 CGS points from your grade**;
- For each subsequent day, up to a maximum of seven days total, **a further one CGS point will be deducted for each day**, or part of a day, up to a maximum of seven days late;
- Over seven days late, **a grade of G3 will be awarded**.

Assignment Information – Late Penalties

It is therefore very important that you submit your assignment before the deadline.

Any time beyond 23:59 on the submission due date will be considered late.

Make sure to leave enough time (> 30 minutes) to upload and submit the assignment itself.

Open Q&A / Discussion